BRAINWARE UNIVERSITY
SCHOOL OF COMPUTATIONAL & APPLIED SCIENCES
DEPARTMENT OF COMPUTATIONAL SCIENCES
BACHELOR OF COMPUTER APPLICATIONS /
BACHELOR OF COMPUTER APPLICATIONS (HONOURS) /
BACHELOR OF COMPUTER APPLICATIONS (HONOURS with RESEARCH) – 2024
SEMESTER – IV

Course Code: BCA47111
Course Name: Full-Stack Development-I
Weekly Contact Hours: 2L+ 4P
Credits: 4
Total Allotted Hours: 30L+ 60P

Course Objective: This course aims to equip students with a comprehensive understanding of Node.js and its practical applications. Students will develop the fundamental concepts of Node.js, including its definition, advantages, and real-world use cases.

Pre-requisite(s): Knowledge of basic web technologies such as HTML, CSS, JavaScript.

Course Outcomes (COs): After completion of this course, students will be able to:

CO1: Understand and evaluate Node.js based web application.
CO2: Apply and assess Node.js module along with NPM and package.json file.
CO3: Create and operate a basic web server with HTTP module and handle requests/responses, and perform essential file system operations.
CO4: Employ advanced debugging with Node.js debugger and develop/compose Visual Studio Code for event-driven programming, and configure routes in Express.js.
CO5: Develop a functional web application and employ Node.js and MongoDB, showcasing proficiency in CRUD operations, connection management, and middleware integration.

Module 1: Introduction to Node-JS and Setup Development Environment **[4H(L)]**
Definition and usage of Node.js, advantages of Node.js, use cases and real-world examples, traditional web server model vs Node.js process model, installing Node.js on Windows / MacOS, setting up a basic development environment.

Practical: **[6H(P)]**
Working in REPL (Node.js interactive shell), Node.js console and basic commands.

Module 2: Node JS Modules and Node Package Manager **[4H(L)]**
Functions in JavaScript, introduction to buffers, Node.js modules, core modules, local modules, require (). Overview of the Node Package Manager (NPM), installing packages locally and globally, managing dependencies with package.json, updating packages.

Practical: **[12H(P)]**
Implementation of functions in JavaScript, Node.js modules, core modules, local modules, require (), installing packages locally and globally, managing dependencies with package.json, updating packages.

Module 3: Creating a Basic Web Server using HTTP Module **[6H(L)]**
Handling HTTP requests and responses, sending requests to the server, Introduction to the file system module (fs), Reading and writing files synchronously and asynchronously, opening, updating, and deleting files, other input/output

operations.

Practical: **[14H(P)]**
Working with handling HTTP requests and responses, sending requests to the server, Introduction to the file system module (fs), reading and writing files synchronously and asynchronously, opening, updating, and deleting files, other input/output operations.

Module 4: Debugging Node.js Application, Events, Express.js **[6H(L)]**
Core Node.js Debugger, debugging with Visual Studio code, Understanding and utilizing EventEmitter class, returning event emitters from functions, inheriting events in Node.js, Configuring routes in Express.js, Building a basic web application with Express.js, middleware in Express.js.

Practical: **[12H(P)]**
Working with Node.js applications. Configuring routes in Express.js and events.

Module 5: Serving Static Resources and Database Connectivity **[10H(L)]**
Serving static files with Express.js, working with middleware in Express.js, NoSQL, Introduction to MongoDB and its use cases, connecting to MongoDB from Node.js, CRUD operations with MongoDB, reading data with the select command, updating records, deleting records, Managing connection strings and configurations.

Practical: **[16H(P)]**
Implementation of serving static files with Express.js, working with middleware in Express.js, NoSQL, connecting to MongoDB from Node.js, CRUD operations with MongoDB, reading data with the select command, updating records, deleting records, Managing connection strings and configurations.

Course Outcomes (COs) and Modules Mapping:

Sl. No.	Course Outcome (COs)	Mapped Module(s)	Learning levels (as per Bloom's Taxonomy from K1 to K6)
1.	CO1	M1	K2, K5
2.	CO2	M2	K3,K5
3.	CO3	M3	K3, K6
4.	CO4	M4	K3,K4
5.	CO5	M5	K3, K6

Text books:
1. Node.js, MongoDB and Angular Web Development, Brad Dayley, Brendan Dayley, Caleb Dayley, Addison-Wesley,2nd Edition.

Reference books:
1. Full Stack Development with MongoDB, Sharma, Manu. BPB Publications, 2022.
2. Node.js in Action, Alex Young, Bradley Meck, and Mike Cantelon, Manning,2nd Edition
3. Node.js by Example, Tsonev, Packt Pub Ltd.